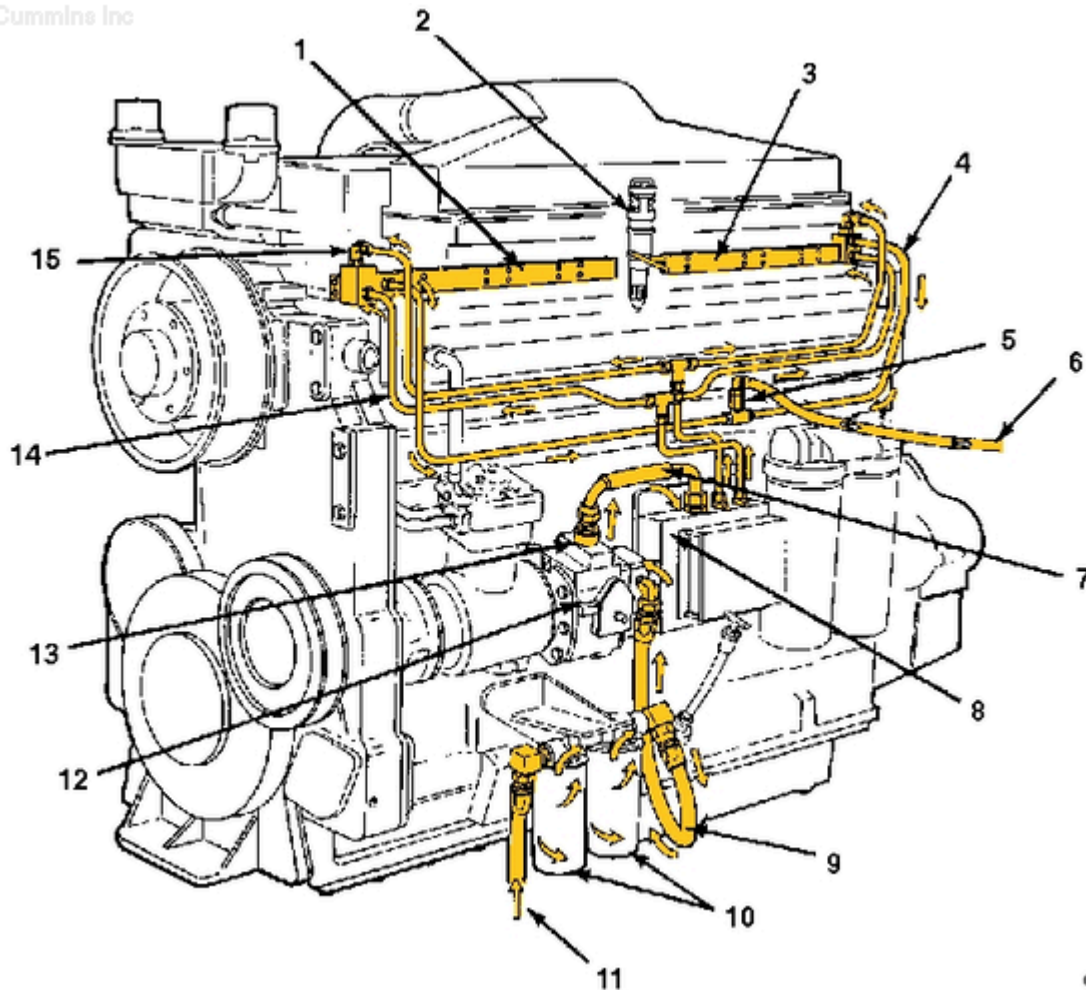


Flow Diagram

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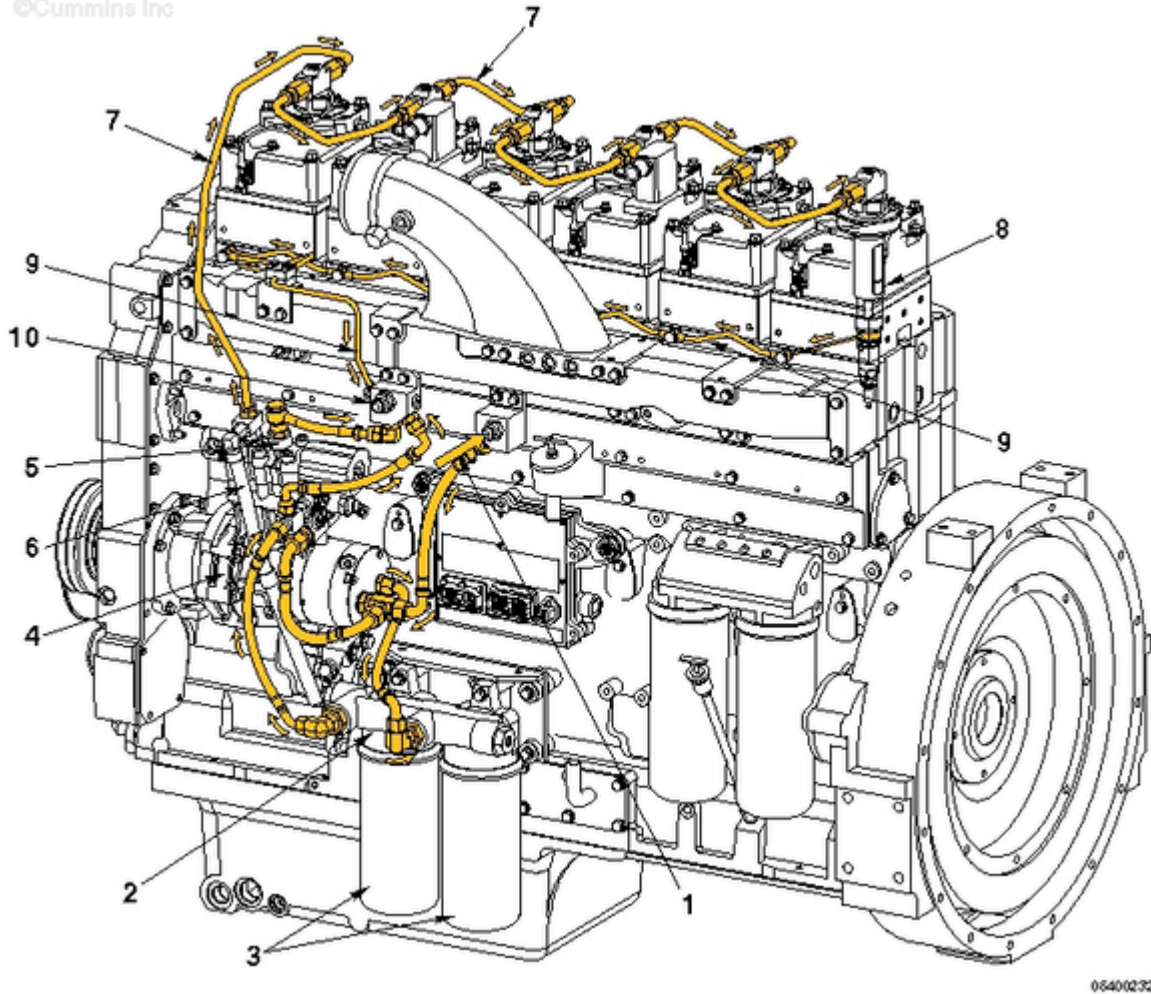


QSK19 with Mechanically Actuated Injector

1. Fuel manifold (front)
2. Injector
3. Fuel manifold (rear)
4. Fuel drain line
5. Fuel drain line check valve
6. Fuel drain line (engine to fuel tank)
7. Fuel control valve supply line
8. Electronic fuel control valve assembly
9. Fuel inlet line
10. Fuel filters with water separation
11. Fuel inlet line (tank to fuel filters)
12. Fuel pump

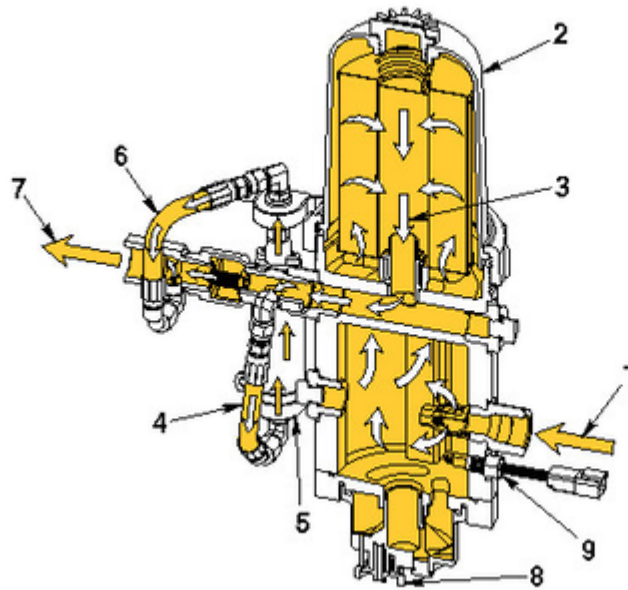
13. Fuel supply check valve (integral to fitting)
14. Fuel rail supply line
15. Fuel timing supply line.

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QSK19 with Electronically Actuated Injector

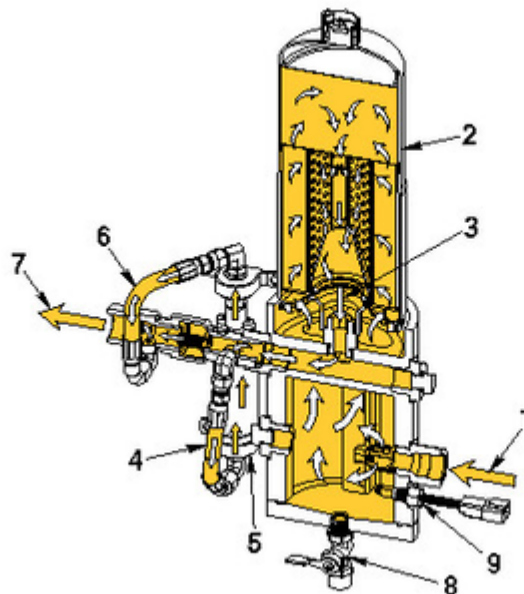
1. Fuel supply from geroter
2. Fuel filter head assembly
3. Second stage fuel filters
4. Fuel pump
5. Mechanical dump valve
6. Vent for injector supply lines
7. Injector supply line
8. Injector
9. Fuel return line
10. Air bleed valve and fuel return to original equipment manufacturer (OEM) fuel tank.



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QSK19 MCRS Industrial Pro™ Stage 1 Single Flow Fuel Filter Head

1. Fuel inlet from tank
2. Filter
3. Fuel from filter to manifold
4. Lift pump suction
5. Lift pump
6. Lift pump discharge
7. Filter outlet to gerotor
8. Water drain
9. Water in fuel sensor.



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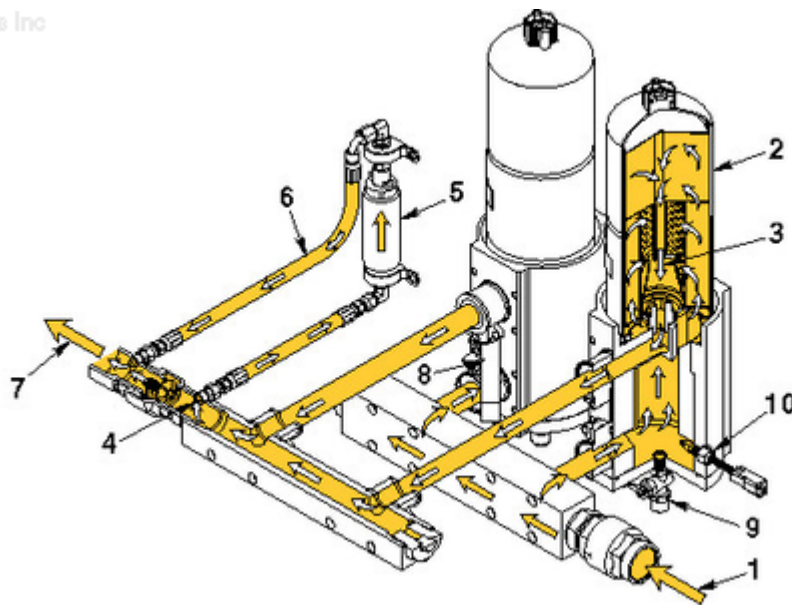
QSK19 MCRS Marine Sea Pro™ 5 Stage 1 Single Flow Fuel Filter Head

Stage 0 Filter - Marine applications specify a primary, off engine filtration system that is supplied by the customer. This is the first filter system after the fuel storage tank and is referred to as the Stage 0 filtration system. The Stage 0 filter system **must** include a water separator and typically

uses a more coarse media (10 micron maximum) with a larger capacity as compared to the Stage 1 (7 micron) and Stage 2 (3 micron) filter systems. A fuel centrifuge is a common acceptable alternative to element type filter media for Stage 0 filtration.

1. Fuel inlet from Stage 0 filter
2. Filter
3. Fuel from filter to manifold
4. Lift pump suction
5. Lift pump
6. Lift pump discharge
7. Filter outlet to gerotor
8. Drain valve
9. Water-in-fuel (WIF) sensor.

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QSK19 MCRS Marine Sea Pro™ 5 Stage 1 Duplex Fuel Filter Head

1. Fuel inlet from Stage 0 filter
2. Filter
3. Fuel from filter to manifold
4. Lift pump suction
5. Lift pump
6. Lift pump discharge
7. Filter outlet to gerotor
8. On/off valves
9. Drain valve
10. WIF sensor.

